Defence Guide · Student 3

Student 3: Discussion & Conclusions Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

> **Your role.** You close the defence. The committee's last impression is yours: finish calmly on the conclusion and the clinical message.

You present	Slides 24–33 (≈ 4 minutes)
You own in the thesis	Chapter 4 Discussion (4.1–4.8), Chapter 5 Conclusions & Recommendations
Your handover line	“Thank you for listening; we welcome your questions.” (leave the r = 0.47 slide on screen)
Your teammates	Student 1: Introduction & Methods (slides 1–12) · Student 2: Results (slides 13–23)

What is in this guide

1. The whole study: two-minute script, core facts, terms, “never say” list (same in all three guides)
2. Your chapter, section by section: what it says, the numbers, the sources, what to watch
3. Your key sources: each reference number in your chapter and what it found
4. Your slides, one by one: what is on screen, your script, timing and traps
5. Questions: on your part, on the whole study, and difficult ones, with model answers
6. Editing your chapter safely and in your own words
7. Your 14-day plan
8. Self-test with answers
9. Checklist for the day before

Part 1 · The whole study (everyone must know this)

> **The committee can ask any of you about any part.** You split the presentation, not the knowledge. Learn this part so well that you can explain the whole study without slides.

The study in two minutes (practise saying this out loud)

"Many people now look up their symptoms online before seeing a doctor, and more and more of them ask AI chatbots like ChatGPT. For some people this searching makes them more worried, not less. That is called cyberchondria, and it is closely linked to health anxiety. No one had studied this in Baghdad, and AI chatbots had hardly been studied next to search engines. We ran an online survey of 300 adults in Baghdad between 17 July 2026 and 11 September 2026, using two established scales: the SHAI-14 for health anxiety and the CSS-12 for cyberchondria. We found that AI chatbots are now the main source of health information: 78.0% ticked them, more than search engines (56.3%). About one in five (19%) had elevated health anxiety. Health anxiety and cyberchondria were moderately linked, r = 0.47, and this held after adjusting for everything else. The link ran mainly through excessive searching and post-search distress, which fits a vicious cycle, although one survey cannot prove direction. We found no reliable evidence that AI itself adds to cyberchondria beyond health anxiety. And searching shapes the consultation: about a third delay care because they feel reassured, and about a third never tell their doctor what they searched. So our main message for clinicians is simple: ask patients, without judgement, what they searched or asked an AI."

Core facts (memorise; examiners test these)

Topic	Fact
Design	Analytical cross-sectional online survey; STROBE and CHERRIES checklists
Who and when	**300** adults, Baghdad, 17 July 2026 – 11 September 2026; convenience + snowball sampling
Sample size	**194** needed for r = 0.20 (α 0.05, 80% power); margin of error ±5.7%
Sample profile	62.3% aged 20–29 · 56.3% female · **44.0%** medical students · 63.3% health background · 21.7% chronic condition · 70.7% answered in Arabic
Tools used	AI chatbots **78.0%** ticked (88.7% incl. named) · search engines 56.3% · medical platforms 18.3% · social media 10.7% · ChatGPT 59.3% · AI without any search engine 41.0%
Motives	convenience 80.0% · "do I need a doctor?" 57.7% · instant reassurance 51.0%
Health anxiety	SHAI mean 12.3 ± 5.8 (0–42) · **19%** elevated (≥ 18; 95% CI 15.0–23.8%) · 0.7% at ≥ 29
Cyberchondria	CSS mean 28.9 ± 8.4 (12–60) · excessiveness highest, compulsion lowest
Reliability	α 0.81 (SHAI), 0.86 (CSS); ω 0.82, 0.87 · CSS-12 4-factor CFA: CFI 0.95, RMSEA 0.070, SRMR 0.053
Arabic vs English	CSS-12 fully (scalar) invariant; SHAI-14 metric invariant; no item DIF after FDR
Main result	**r = 0.47** (95% CI 0.38–0.55); explains 22% of the variance · partial r = 0.48
By facet	excessiveness 0.46 · distress 0.45 · compulsion 0.25 · reassurance 0.24
Regression	R^2 0.030 (demographics) → 0.133 (+ digital use) → 0.317 (+ health anxiety, ΔR^2 18.4%); β = 0.48
Who is anxious	AOR **3.12** per SD of cyberchondria · 3.83 reassurance motive · 0.61 per older age group · AUC 0.83 → 0.79
Pathways	via excessive searching 0.16 (CI 0.11–0.23) · via distress 0.22 (direct → 0.01) · reverse 0.15
Behaviour	delay care when reassured **34.7%** · ignore symptom 30.0% · several doctors 25.3% · urgent care 22.0% · self-medication 21.3%
Doctor relationship	never tell the doctor **31.7%** · always tell 27.0% · trust strained 34.7% · higher cyberchondria → more disclosure (OR 0.57, q 0.012)

AI effect	+5.3 CSS points (inclusive) but +1.5, p = 0.31 (strict) → no reliable independent effect · 32 named-not-ticked · 34 non-users
Robustness	every sensitivity analysis left the main result unchanged (Table 13)

Terms you must be able to explain in one sentence

Term	Plain explanation
Health anxiety	Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by reassurance seeking.
Illness anxiety disorder	The DSM-5 diagnosis at the severe end of health anxiety. We did **not** diagnose it; we measured symptoms.
Cyberchondria	Excessive or repeated online health searching that increases rather than relieves distress.
SHAI-14	14 items, each scored 0–3 (total 0–42); higher = more health anxiety. ≥ 18 = elevated (screening threshold).
CSS-12	12 items scored 1–5 (total 12–60) with four 3-item facets: excessiveness, compulsion, distress, reassurance seeking.
Cross-sectional	Everything measured once, at one time point: shows associations, not cause and effect.
Convenience / snowball sampling	People who were easy to reach, who then shared the link with others. Quick, but not representative.
Mean ± SD	The average score and how spread out the scores are.
Median (IQR)	The middle score and the range of the middle half; used when data are skewed.
p-value	How surprising the result would be if there were truly no effect; $p < 0.05$ is called significant.
95% CI	The range in which the true value probably lies; narrower = more precise.
Correlation r	How closely two scores move together (0 = none, 1 = perfect). 0.47 is moderate.
Partial correlation	The same link after removing the effect of other variables.
Mann–Whitney / Kruskal–Wallis	Non-parametric tests comparing two / three or more groups, used because scores were not normally distributed.
χ^2 test	Tests whether two categorical variables are related (e.g., social-media use and elevated health anxiety).
FDR (q-value)	Correction for running many tests; a result is "robust" if $q < 0.05$, otherwise exploratory.
Linear regression, B and β	Predicts a score from several factors at once. B = change in CSS points per unit; β = the same on a standardised scale (to compare factors).
R^2 and ΔR^2	Share of variation explained by the model; ΔR^2 = extra share explained by adding a block of variables.
Logistic regression, AOR	Predicts a yes/no outcome. Adjusted odds ratio: how many times the odds change, holding other factors constant.
AUC	How well a model separates two groups (0.5 = chance, 1 = perfect).
Mediation	Splitting a link into a part that runs through a third variable (indirect) and the rest (direct). Statistical, not causal.
Moderation	Whether a third variable changes the strength of a link (e.g., does AI use make the link stronger? No).
Cronbach α / McDonald ω	Internal consistency: do the items of a scale hang together? ≥ 0.7 is acceptable, ≥ 0.8 good.
CFA (CFI, RMSEA, SRMR)	Checks the items group into the intended factors. Good: CFI ≈ 0.95+, RMSEA ≤ 0.08, SRMR ≤ 0.08.
Measurement invariance	Checks the scale measures the same thing in two groups (here Arabic vs English) before comparing them.

Sensitivity analysis	Re-running the main analysis in a different way to see if the result holds.

Never say (overclaims the committee will catch):

- "Health anxiety *causes* cyberchondria" → say "is associated with", "is compatible with a cycle".
- "19% *have* health anxiety disorder" → "19% scored in the elevated range on a screening tool".
- "AI makes people anxious" → "we found no reliable independent effect of AI use".
- "Our results represent Baghdad" → "a young, online, convenience sample; associations were consistent across subgroups".
- "The Arabic version is validated" → "we tested it: equal reliability and formal invariance, but no formal back-translation".

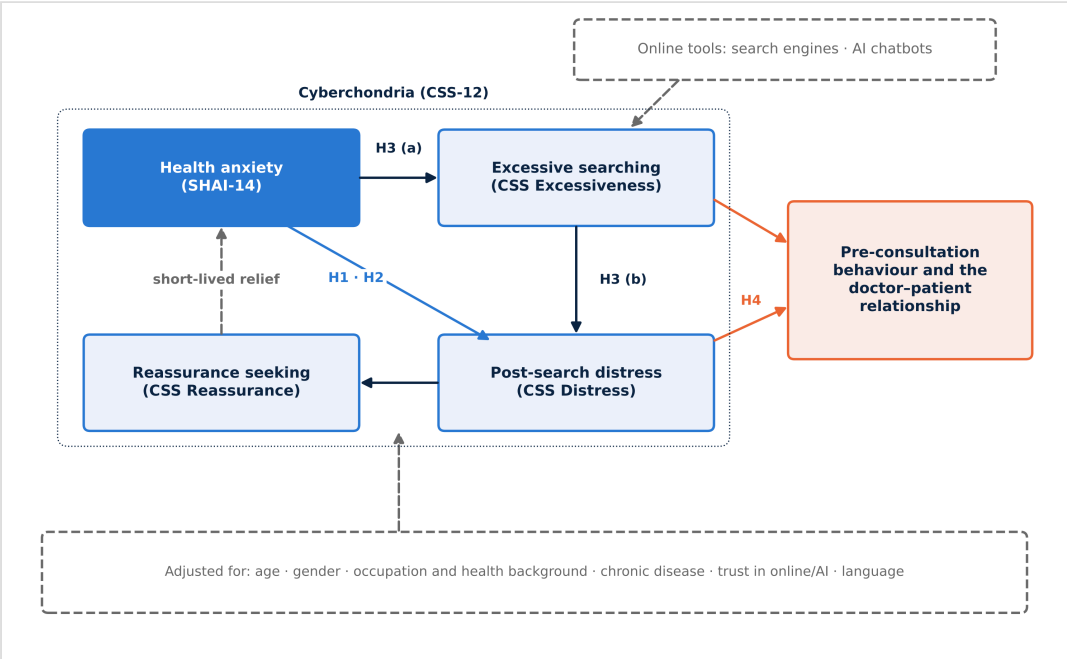


Figure 1 of the thesis: the conceptual framework. Learn to explain every arrow.

Part 2 · Your chapter, section by section

For each section: its **purpose**, the **key points and numbers** (with the reference numbers used in the thesis), and what to **watch**.

4.1 Principal findings

The four headline findings.

- First CSS-12 study in Baghdad; among the first to put AI chatbots next to search engines.
- (1) AI used by 78.0%–88.7%; (2) r = 0.47, independent; (3) concentrated in excessiveness and distress; (4) searching shapes the consultation.

Watch: This is your opening slide content (slide 25).

4.2 Health anxiety and cyberchondria

Why our r is weaker than meta-analyses.

- Ours 0.47 vs 0.62–0.63 [18][19].
- Reasons: measurement error (corrected ≈ 0.56); low, narrow health anxiety (mean 12.3); online recruitment of confident users; CSS-12 instead of the 33-item CSS.
- 19% elevated is a screening estimate; 7%–33% across cut-offs; consistent with high pandemic health anxiety in Iraq [9].

Watch: Learn the four reasons as a numbered list.

4.3 Do they fuel each other?

- Consistent with the cognitive model and the Dutch longitudinal study [5][21].
- But cross-sectional mediation is biased for longitudinal processes [54]; an indirect effect cannot identify the true order [55]; shared method variance [56].
- H3 supported statistically, not causally.

Watch: If you remember one sentence: “compatible with a cycle, not proof of direction”.

4.4 AI chatbots

- Our 78–89% vs ≈ 20% in a 2024 US survey [28] and 21% vs 98% search engines in 2025 [57].
- Young sample: 96% at 18–19 y vs 72% at ≥ 40 y.
- Inclusive AI effect not robust; no moderation; absence of evidence, not evidence of absence.
- Trust → fewer multiple-doctor visits but a trend to reassured delay (OR 1.33, p 0.088), echoing MENA data [36].

4.5 The consultation

- Reassured delay (34.7%): saves visits for minor symptoms, risks delayed diagnosis; occurs at all anxiety levels.
- Cyberchondria with urgent care (exploratory), mirroring healthcare use [6][16].
- Non-disclosure 31.7%; higher cyberchondria → more disclosure (robust); health anxiety → concealment (exploratory); discussed searches strengthen the relationship [33].
- H4 partially supported.

4.6 Medical students

- Higher unadjusted health anxiety disappeared after adjustment (age). SHAI–CSS r 0.57 in medical students vs 0.40 others.
- “Medical student syndrome”: weak evidence; a Palestinian study with the same SHAI ≥ 18 found medical students lowest [58].
- Medical knowledge did not protect against cyberchondria.

Watch: Expect a question here: you are medical students yourselves.

4.7 Regional comparison

- Jeddah: common, linked with smartphone addiction [34]; Saudi primary care: linked with health anxiety [35]; Lebanon: Arabic CSS-12 structure similar [15].
- Egypt: 1435 medical students, 85.8% moderate cyberchondria [59]; Saudi CSS-12 validation [46].

4.8 Strengths and limitations

- Strengths: established scales, bilingual with invariance testing, AI separate, complete data, FDR, robust and bootstrap methods, sensitivity analyses, links to behaviour.
- Limitations: cross-sectional; convenience sample (young, 44% medical students); Arabic wording without back-translation; single-item behaviour measures; self-report; many tests and small subgroups; AI classification; careless responding.

Watch: For each limitation, know what was done about it (Part 5).

5.1 Conclusions (six)

- AI chatbots a leading source (78.0% / 88.7%).
- 19% elevated (screening; 0.7% at 29).
- r = 0.47, strongest predictor.
- Compatible with a cycle, direction unproven.
- No reliable independent AI effect.
- Searching shapes the consultation: delay 34.7%, non-disclosure 31.7%, strained trust 34.7%.

5.2 Recommendations

- Clinicians: ask non-judgementally; recognise repeated searching as a sign; screen with SHAI; CBT strategies; teach digital/AI literacy in medical school.
- Ministry of Health: Arabic digital and AI health-literacy campaigns; reliable Arabic triage sources.
- AI developers: anxiety-aware answers, no escalation to rare diagnoses, Arabic.
- Research: interventions (only 9 registered cyberchondria studies, none in Iraq [60]); longitudinal and experimental designs; probability sampling; adapted Arabic scales.

Watch: Recommendations must follow from results: be ready to link each to a finding.

Part 3 · Your key sources

Every reference cited in your chapter, with its number in the thesis. Know the bold ones well: examiners often ask “who found that?”.

No.	Reference (as listed in Chapter 6)
[5]	Salkovskis PM, Warwick HMC. Morbid preoccupations, health anxiety and reassurance: a cognitive-behavioural approach to hypochondriasis. Behav Res Ther. 1986;24(5):597-602. doi:10.1016/0005-7967(86)90041-0
[6]	Tyrer P. Recent advances in the understanding and treatment of health anxiety. Curr Psychiatry Rep. 2018;20(7):49. doi:10.1007/s11920-018-0912-0
[9]	Karim SK, Taha PH, Amin NMM, Ahmed HS, Yousif MK, Hallumy AM. COVID-19-related anxiety disorder in Iraq during the pandemic: an online cross-sectional study. Middle East Curr Psychiatry. 2020;27(1):55. doi:10.1186/s43045-020-00067…
[15]	Hallit S, Rogoza R, Abi Semaan C, Azzi V, Sawma T, Obeid S. Validation of the Arabic version of the cyberchondria severity scale 12 items (CSS-12-Ar) among a sample of Lebanese adults. BMC Psychiatry. 2023;23(1):618. doi:10.1186/s…
[16]	Mathes BM, Norr AM, Allan NP, Albanese BJ, Schmidt NB. Cyberchondria: overlap with health anxiety and unique relations with impairment, quality of life, and service utilization. Psychiatry Res. 2018;261:204-11. doi:10.1016/j.psych…
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[21]	te Poel F, Baumgartner SE, Hartmann T, Tanis M. The curious case of cyberchondria: a longitudinal study on the reciprocal relationship between health anxiety and online health information seeking. J Anxiety Disord. 2016;43:32-40. …
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[27]	Ayers JW, Poliak A, Dredze M, Leas EC, Zhu Z, Kelley JB, et al. Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. JAMA Intern Med. 2023;183(6):589-96. doi…
[28]	Ayo-Ajibola O, Davis RJ, Lin ME, Riddell J, Kravitz RL. Characterizing the adoption and experiences of users of artificial intelligence-generated health information in the United States: cross-sectional questionnaire study. J Med …
[29]	Wetzel AJ, Klemmt M, Müller R, Rieger MA, Joos S, Koch R. Only the anxious ones? Identifying characteristics of symptom checker app users: a cross-sectional survey. BMC Med Inform Decis Mak. 2024;24(1):21. doi:10.1186/s12911-024-0…
[30]	Svestkova A, Huang Y, Smahel D. Factors that influence trust and willingness to use generative AI for health information: a cross-sectional study. Digit Health. 2025;11:20552076251360973. doi:10.1177/20552076251360973
[31]	Dığış M, Kaya K, Demir B, Çiçek İN, Akdağ D, Akdağ Ş. Frequency of use and attitudes toward AI-based health counseling among patients attending the dermatology outpatient clinic: a cross-sectional study. Cutan Ocul Toxicol. 2026 A…
[33]	Luo A, Qin L, Yuan Y, Yang Z, Liu F, Huang P, et al. The effect of online health information seeking on physician-patient relationships: systematic review. J Med Internet Res. 2022;24(2):e23354. doi:10.2196/23354
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[36]	Dahshan A, Mahfouz Khalil MI, Abdelfatah D, Abu Ejheisheh M, Alshammari MSH, Mekdad AK, et al. Influence of online health information and artificial intelligence on decision-making among migraine patients: a multinational cross-se…
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[44]	Alshayea AK. Latent structure, measurement invariance, and reliability of an Arabic version of the Short Health Anxiety Inventory. J Exp Psychopathol. 2020;11(2). doi:10.1177/2043808720912629
[46]	Alshaibani M. Validity and reliability of the short form of Cyberchondria Severity Scale (CSS-12) in Saudi Arabia. J Consum Health Internet. 2023;27(3):282-93. doi:10.1080/15398285.2023.2237347
[54]	O'Laughlin KD, Martin MJ, Ferrer E. Cross-sectional analysis of longitudinal mediation processes. Multivariate Behav Res. 2018;53(3):375-402. doi:10.1080/00273171.2018.1454822
[55]	Rohrer JM, Hünermund P, Arslan RC, Elson M. That's a lot to process! Pitfalls of popular path models. Adv Methods Pract Psychol Sci. 2022;5(2). doi:10.1177/25152459221095827
[56]	Podsakoff PM, Podsakoff NP, Williams LJ, Huang C, Yang J. Common method bias: it's bad, it's complex, it's widespread, and it's not easy to fix. Annu Rev Organ Psychol Organ Behav. 2024;11:17-61. doi:10.1146/annurev-orgpsych-11072...
[57]	Yun HS, Bickmore T. Online health information-seeking in the era of large language models: cross-sectional web-based survey study. J Med Internet Res. 2025;27:e68560. doi:10.2196/68560
[58]	Fasfoos A, Sunkrot MS, Jadallah I, Id'eas AH, Hoshieh M, Haytham T, et al. Health anxiety among medical and non-medical university students in Palestine: associations with intolerance of uncertainty, daily phone use, and field of ...
[59]	El-Zoghby SM, Zaghloul NM, Tawfik AM, Elsherbiny NM, Shehata SA, Soltan EM. Cyberchondria and smartphone addiction: a correlation survey among undergraduate medical students in Egypt. J Egypt Public Health Assoc. 2024;99(1):7. doi...
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Part 4 · Your slides, one by one

The script below is the speaker-notes text of your slides in the presentation. Practise until you can say it in your own words without reading.

Slide 24 · Discussion

Thank you. I will discuss what these results mean, how they compare with other studies, our limitations, and our conclusions.

Slide 25 · n = 300 · r = 0.47

To our knowledge, this is the first study to use the CSS-12 in Baghdad, and among the first anywhere to look at AI chatbots next to search engines before seeing a doctor. Our main finding, r = 0.47, supports our first two hypotheses: health anxiety and cyberchondria go together, independently of everything else we measured.

Slide 26 · Comparison with Worldwide Results

How do we compare with the world? Our correlation of 0.47 is weaker than the pooled estimates of about 0.62 from meta-analyses. Several reasons: first, measurement error; corrected for reliability it would be about 0.56, close to the meta-analytic range. Second, our sample had low health anxiety, and a narrow range weakens correlations. Third, online recruitment may favour confident, habitual internet users. Fourth, most earlier studies used the full 33-item scale.

Slide 27 · Comparison with Regional Studies

Our findings agree with the Arab region. In Jeddah, cyberchondria was common and linked with smartphone addiction. In Saudi primary care, cyberchondria was linked with health anxiety. In Lebanon, the Arabic CSS-12 had the same structure as ours. And among 1435 Egyptian medical students, the group closest to ours, most had moderate cyberchondria.

Slide 28 · AI Chatbots & the Consultation

AI use in our sample, 78.0% to 88.7%, is far higher than in a 2024 US survey, about 1 in 5, and a 2025 survey, 21%. Our sample is young; AI use fell from 96% at 18 to 19 years to 72% at 40 and above. On the consultation: delaying care because of reassurance can save visits for minor symptoms, but risks delayed diagnosis for serious ones, and it happened across all anxiety levels. About a third never tell their doctor what they searched. So doctors should ask.

Slide 29 · 44.0% of our sample were medical students, a distinct group:

44.0% of our sample were medical students, a distinct group: high medical awareness, academic stress, and exposure to descriptions of serious disease. They had higher unadjusted health anxiety, but this disappeared after adjustment, because they are younger, and age was the independent factor. Their cyberchondria scores were identical to everyone else's. So medical knowledge did not protect against cyberchondria.

Slide 30 · Limitations

We have several limitations. First, the design is cross-sectional, so we cannot prove direction; the mediation is statistical, not causal. Second, the sample is online convenience and snowball, young, and heavy in medical students, so percentages cannot be generalised to all of Baghdad, although associations were consistent across subgroups. Third, the Arabic wording was ours, without formal back-translation, although reliability and means were the same in both languages. Fourth, behaviour items were single, non-validated questions. Fifth, many tests; we corrected with FDR.

Slide 31 · Future Directions & Recommendations

For the future: longitudinal studies to test the cycle over time; experiments comparing AI chatbot and search-engine answers; probability sampling of all Baghdad adults; and formally adapted Arabic scales. ClinicalTrials.gov lists only nine cyberchondria studies and none in Iraq, so intervention trials are a clear gap. For practice: doctors should ask about searches; the Ministry may consider digital and AI health-literacy campaigns; and AI developers may consider anxiety-aware answers.

Slide 32 · Conclusion

To conclude. AI chatbots, mainly ChatGPT, are now the main source of health information before seeing a doctor in our sample. About one in five has elevated health anxiety. Health anxiety and cyberchondria are moderately and robustly linked, r = 0.47, compatible with a cycle. We found no reliable independent effect of AI. And searching shapes the consultation: delays, non-disclosure and strained trust. Thank you.

Slide 33 · r = 0.47

Leave this slide up during questions. Each student answers questions on their own part; Student 2 takes statistics questions.

Delivery for your part: ≈ 4 minutes; about 25–35 seconds per content slide and 10 seconds per divider. Point at the number you are saying. Pause after each key number. End with: “Thank you for listening; we welcome your questions.” (leave the r = 0.47 slide on screen)

Part 5 · Questions and model answers

A. Questions on your part

Q: Why is your correlation weaker than other studies?

Four reasons: measurement error (corrected r ≈ 0.56), low and narrow health anxiety, online recruitment of confident users, and the short CSS-12.

Q: Is your sample representative of Baghdad?

No: young, educated, many medical students. We do not generalise percentages; the associations were consistent across subgroups.

Q: What is new in your study?

First CSS-12 data from Baghdad; AI chatbots separated from search engines; psychology linked to behaviour before the consultation.

Q: Why is AI use so high compared with the US?

A young, educated, health-oriented sample; AI use fell clearly with age (96% → 72%). It is not an estimate for all adults.

Q: So does AI make cyberchondria worse?

We found no reliable independent effect beyond health anxiety. That is absence of evidence, not evidence of absence; experiments are needed.

Q: Is delaying care because of reassurance good or bad?

Both: it may prevent unnecessary visits for minor symptoms but risks missing serious ones. It occurred at all anxiety levels.

Q: Why do people with more cyberchondria disclose more?

Searching is central to their behaviour and they bring findings as part of reassurance seeking; health-anxious patients may hide searches for fear of being seen as “worried well” (exploratory).

Q: Are medical students more anxious?

Unadjusted yes, but it disappeared after adjustment (younger age). Their cyberchondria was the same as others'. A Palestinian study [58] found medical students lowest.

Q: What is your biggest limitation?

Cross-sectional design: no causal direction. Second, the convenience sample.

Q: If you repeated the study, what would you change?

Probability sampling across ages, formal Arabic adaptation (or the published versions), a clearer AI-use question, and follow-up over time.

Q: How do your findings compare with the Arab region?

Consistent: Jeddah, Saudi primary care and Lebanon all link cyberchondria with health anxiety; Egyptian medical students show high cyberchondria.

Q: What should a doctor do differently on Monday?

Ask every patient, without judgement, what they searched or asked an AI; treat repeated searching and reassurance seeking as signs of health anxiety; avoid endless reassurance.

Q: What should the Ministry do?

Support Arabic digital and AI health-literacy campaigns and reliable Arabic triage sources, warning especially about false reassurance.

Q: Why recommend intervention trials?

ClinicalTrials.gov lists only nine cyberchondria studies, none in Iraq; brief Arabic digital or CBT interventions are a clear gap.

Q: Was H4 supported?

Partially: only higher cyberchondria → more disclosure survived FDR; urgent care and strained relationship were exploratory.

Q: How do you know the results are not due to careless responders?

We excluded straight-liners and annotated responses in sensitivity analyses; the correlation did not change (Table 13).

Q: What does "absence of evidence is not evidence of absence" mean here?

Not finding a robust AI effect does not prove AI has no effect; our design and the small non-user group could miss it.

Q: Why is the Discussion careful with words like "cycle"?

Because one survey cannot show direction; we use "compatible with" to stay within what the data show.

B. Questions anyone can be asked (know the answer even if it is not your part)

Q: Summarise your study in one minute.

Use the two-minute script in Part 1, shortened: design, sample, two scales, r = 0.47, AI result, behaviour, clinical message.

Q: What is your main finding?

Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47), independent of other factors.

Q: What did you find about AI chatbots?

They are the main source (78.0% ticked), but there was no reliable independent effect on cyberchondria beyond health anxiety.

Q: What percentage had elevated health anxiety?

19% at SHAI ≥ 18, a screening threshold, not a diagnosis.

Q: What is your biggest limitation?

The cross-sectional design: no direction of cause.

Q: What is the clinical message?

Ask patients, without judgement, what they searched or asked an AI.

Q: Why did you split the presentation this way?

By chapter, but each of us studied the whole thesis.

Q: Were your hypotheses written before you collected the data?

Answer truthfully. A safe true answer: they follow from our objectives and previous research, and the questionnaire was designed to measure these things; we tested all of them and report every result, including those not supported.

Watch out: Never claim they were pre-registered.

Q: Who did the statistical analysis?

Answer honestly according to what you did. If AI tools helped with code or drafting, say so in line with your college policy and the declaration in the thesis.

C. Difficult or hostile questions

Q: "Your recommendations go beyond your data."

Each one is tied to a finding (e.g., ask about searches ← 31.7% never disclose); policy recommendations are phrased as "may consider".

Q: "Why should we believe a self-report survey?"

Self-report is standard for these constructs; the scales are validated, reliable and invariant; we acknowledge recall and social-desirability bias.

Q: "Isn't the AI finding just a coding decision?"

Yes, and we say so openly; that is why we conclude there is no reliable evidence of an independent AI effect.

D. Limitations and what was done about each (all students)

Limitation	What we did / how to answer
Cross-sectional design	Reverse-direction models; wording "compatible with"; longitudinal studies recommended
Convenience/snowball sample	Percentages not generalised; associations consistent across subgroups
Arabic wording (no back-translation)	Equal reliability and means; CSS-12 fully invariant, SHAI-14 metric; no item DIF
Single-item behaviour measures	Reported descriptively; associations labelled exploratory
Many tests	FDR correction; robust vs exploratory separated
Small subgroups (34 non-users)	Acknowledged; AI effect called "not robust"
Self-report and careless responding	Anonymous; straight-liners and annotated responses excluded in sensitivity analyses (no change)

> **How to answer any question:** pause → repeat the question briefly → answer in two or three sentences → stop. If you do not know: "We did not examine that directly; based on our data we would expect…, and it is a good question for future research." Never invent a number. If a teammate struggles: "If I may add…".

Part 6 · Editing your chapter

> **The Word file is the master copy.** Edit only the .docx, in one shared OneDrive / Word Online file with **Review → Track Changes** on. Keep an untouched copy (Thesis_ORIGINAL.docx). Avoid Google Docs: it breaks the automatic numbering.

For your chapter

- Discussion is where your own interpretation belongs: rewrite each paragraph in your voice, keeping every [n] with its claim.
- Add genuine local insight (how patients in Baghdad use ChatGPT, pharmacies, cost of visits) only if it is true.
- Do not strengthen claims: keep “compatible with”, “exploratory”, “may consider”.
- Recommendations can be personal and practical, but each must follow from a result.
- Keep the limitations complete: removing one makes the thesis look less honest, not stronger.

Never touch

- Numbers, %, p, q, CI, r, OR, β, n
- Reference numbers like [18] (move them with their sentence)
- The reference list
- Table/figure captions, table of contents and lists (automatic fields)
- Heading numbers and styles
- Times New Roman 14 and page numbering

The rewrite method (paragraph by paragraph)

1. Read until you understand it; look up anything unclear.
2. Close it and explain it out loud.
3. Write your version from that explanation.
4. Compare: same facts, numbers and [n]; nothing claimed more strongly.
5. Read aloud; change any phrase you would never say.

After editing

1. Ctrl + A, then F9, to update the table of contents, the lists and the captions.
2. Search for “[” to find placeholders left.
3. Swap chapters with a teammate for checking.
4. Export to PDF and look at every page.

> **About AI use, honestly.** Check your college's AI policy with your supervisor and declare AI assistance if required. A sentence you can adapt: *“The statistical analysis code and an initial draft of this report were prepared with the assistance of an AI tool (Claude, Anthropic). All analyses were checked, and the text was reviewed and rewritten by the authors, who take full responsibility for its content.”* Do not use paraphrasing tools or tricks to fool detectors: they make the text worse and can count as misconduct. Real ownership (your words, your understanding, your edit history) is the best protection.

Part 7 · Your 14-day plan

Day	Task
1	Read the Abstract, 1.8 and Chapter 5. Learn the two-minute script (Part 1).
2	Read the whole thesis once, marking what you cannot explain.
3	Core facts and terms (Part 1). Self-test on core facts.
4	Your chapter, sections in Part 2 (first half).
5	Your chapter, second half; your key sources (Part 3).
6	Start rewriting your chapter (Part 6).
7	Finish rewriting. Self-test (Part 8).
8	Swap chapters with a teammate; check accuracy and voice.
9	Your slides: read the scripts (Part 4), then present them alone 3 times.
10	Questions A–D out loud (Part 5). Record yourself once.
11	Team: full rehearsal with timing and handovers.
12	Team: mock committee; each student asks the others their Part 5 questions.
13	Fix weak answers; second full rehearsal; final Word checks.
14	Day before: checklist (last page); one calm run-through; sleep.

Part 8 · Self-test

Cover the answers, answer out loud, then check. Repeat until all are correct.

#	Question	Answer
1	Our r vs meta-analytic r?	0.47 vs 0.62–0.63
2	Four reasons our r is weaker?	Measurement error, low/narrow health anxiety, online recruitment, short CSS-12
3	Disattenuated correlation?	≈ 0.56
4	AI use in our study vs US 2024?	78.0%–88.7% vs ≈ 20%
5	AI use by age?	96% at 18–19 y vs 72% at ≥ 40 y
6	Percentage delaying care when reassured?	34.7%
7	Percentage never telling the doctor?	31.7%
8	Was H3 supported?	Statistically, not causally
9	Was H4 supported?	Partially (only disclosure survived FDR)
10	SHAI–CSS r in medical students vs others?	0.57 vs 0.40
11	Egyptian study size and finding?	1435 medical students; 85.8% moderate cyberchondria
12	Number of registered cyberchondria trials, and in Iraq?	Nine; none in Iraq
13	Biggest limitation?	Cross-sectional design
14	Prevalence range across cut-offs?	About 7% to 33% (0.7% at 29)
15	The clinical message in one sentence?	Ask every patient, without judgement, what they searched or asked an AI
16	Main correlation (everyone)?	r = 0.47

17	Elevated health anxiety (everyone)?	19%
18	AI chatbots ticked (everyone)?	78.0%
19	Never tell the doctor (everyone)?	31.7%
20	Sample size (everyone)?	300

Part 9 · Day-before checklist

✓	Item
☐	All placeholders filled (names, supervisor, college, logo, degree, year, ethics, Google Forms question, Acknowledgment)
☐	Fields updated (Ctrl + A, F9); PDF checked page by page
☐	AI declaration added if required
☐	Presentation: names on slide 1, ethics number on slide 10; USB + PDF + email copies; clicker tested
☐	I can give the two-minute summary and my slides (24–33) without notes
☐	I have answered every question in Part 5 out loud
☐	I scored 100% on the self-test
☐	Printed thesis copy if required; arrive early; agree who stands where

Generated from the study outputs and the presentation, so every number and slide script here matches the thesis and the deck.